

2.1 Applying Product Laws of Exponents to Algebraic Expressions

Lesson Introduction

Lesson Introduction					
	Benchmark	MA.912.NSO.1.2 Generate equivalent algebraic expressions using the properties of exponents.			Learning Targets I can generate equivalent algebraic expressions using the properties (laws) of exponents to find the product of monomial expressions.
	Benchmark Coherence	Building On			Working Toward
		MA.8.NSO.1.3 Extend previous understanding of the laws of exponents to include integer exponents. Apply the laws of exponents to evaluate numerical expressions and generate equivalent numerical expressions, limited to integer exponents and rational number bases, with procedural fluency. MA.8.AR.1.1 Apply the Laws of Exponents to generate equivalent algebraic expressions, limited to integer exponents and monomial bases.			MA.912.NSO.1.3 Generate equivalent algebraic expressions involving radicals or rational exponents using the properties of exponents.
	Essential Questions	- How is an equivalent expression created using the product of powers property? - How is an equivalent expression created using the power of a product property? - How is an equivalent expression created using the power of a power property? - How is the order of operations used when writing an equivalent expression using the laws of exponents?			
	Lesson Narrative	In this lesson, students will determine an equivalent expression using the product laws of exponents. Students will be positioned to use the product of powers law, the power of a power law, and the power of a product law.			
	Component Summary	2.1.1 Warm-Up (~5 min.)	Reflecting on a growth mindset (SE p. 57)	2.1.4 Collaboration (10 min.)	Applying the power of a power law and power of a product law (SE p. 59)
		2.1.2 Guided Instruction (10 min.)	Applying the product of powers law (SE p. 58)	2.1.5 Your Turn! (10-15 min.)	Generating equivalent monomial expressions using the product laws of exponents (SE p. 60)
		2.1.3 Guided Practice (5-10 min.)	Using the product of powers law to generate an equivalent expression (SE p. 59)	2.1.6 Wrap-Up (~5 min.)	Generating equivalent monomial expressions using the product laws of exponents (SE p. 61)
	Resources	Glossary		Materials	Additional Preparation
		<u>Key Terms:</u> - Product of powers - Power of a power - Power of a product <u>Additional Terms:</u> - Exponent - Base - Coefficient		(Optional) Calculator	B.E.S.T. uses “properties” and “laws” interchangeably. Both laws of exponents and properties of exponents need to be used during instruction to ensure students understand they are the same.

 Teacher Tips	Common Misconceptions - Students may have forgotten the laws of exponents from the previous unit. - Students may want to raise a coefficient to the power that an attached variable is being raised.	Things to Consider - Since students may not remember the laws of exponents, this lesson uses the expanded form along with the laws to position them to understand operations with exponents. - This is the first time the term “monomial” is being introduced. - In 8th grade students would have seen some of these expressions, but for Algebra 1 students in 8th grade, this will be their first time seeing this content.
	Literacy and Language Routines Think-Pair-Share Utilize the structure of the 2.1.2 Guided Instruction component. Students are positioned to engage in a collaborative discussion with their partner on equivalent expressions. Consider using the Think-Pair-Share routine to provide students multiple attempts to make sense both independently and through partner discussion for increased precision.	Mathematical Thinking and Reasoning (MTR) Standard MA.K12.MTR.5.1: Patterns and Structure Students will be positioned to look for patterns when learning to write equivalent expressions. In the 2.1.2 Guided Instruction, students will use the product of powers law and expanded form of the monomials to determine what is happening to the coefficients, exponents, and bases when two monomials are multiplied.
	 Differentiate Instruction	
Lesson Notes:		

2.1.1 Warm-Up: Growth Mindset

Component Overview: Students answer questions 1 and 2 independently as they reflect on how they feel about making mistakes.



2.1.1 Warm-Up: Growth Mindset

1. Write a check mark ✓ in the box that best describes how you feel about mistakes.

- ☐ I do not like to admit I made a mistake.
- ☐ I think all mistakes are bad.
- ☐ Mistakes tell me that I won't ever understand.
- ☐ Mistakes make me wonder what I am missing.
- ☐ I think a mistake is a chance to try another strategy.
- ☐ Mistakes help me learn.

Answers will vary.

2. Consider the acronym shown.

How does this interpretation compare to your feelings about making mistakes?

Answers will vary.

Means
I
Start
To
Acquire
Knowledge
Experience
Skills



If time permits, consider asking the following questions to facilitate a discussion about making mistakes.

- *What are the pros and cons of making mistakes?*
- *Why do you think making a mistake is often seen as negative?*
- *How could seeing mistakes as learning opportunities be helpful?*

2.1.2 Guided Instruction: Equivalent Algebraic Expressions

Component Overview: Students will be introduced to generating equivalent algebraic expressions through the properties of exponents. Lead students through the Guided Instruction using whole-group instruction, partner work, and individual work. Use teacher think-aloud strategies for question 2 to model multiplying monomials.



2.1.2 Guided Instruction: Equivalent Algebraic Expressions

The laws of exponents can be applied to generate equivalent algebraic expressions.

- The table shows three expressions and two different methods that can be used to determine an equivalent expression.

Expression	Expanded Form	Product of Powers Law	Equivalent Expression
$x^2 \cdot x^3$	$x \cdot x \cdot x \cdot x \cdot x$	x^{2+3}	x^5
$-4x^2 \cdot 5x^3$	$-4 \cdot x \cdot x \cdot 5 \cdot x \cdot x \cdot x$	$-4 \cdot 5 \cdot x^{2+3}$	$-20x^5$
$6x^2y^0 \cdot -xy$	$6 \cdot x \cdot x \cdot 1 \cdot -1 \cdot x \cdot y$	$6 \cdot -1 \cdot x^{2+1} \cdot y^1$	$-6x^3y$

- Discuss with your partner any patterns you notice between the expanded form and applying the product of powers law.

Answers will vary. I noticed that the value of a variable's exponent in the product of powers law is the same as how many of those variables appear in the expanded form. I noticed that the coefficients get multiplied together as their own step.

- Complete the statements.
When multiplying monomial terms, the coefficients

of the terms can be ☐ added ☒ multiplied to find the

coefficient of the equivalent expression. The

product of powers law of exponents can be

applied to ☐ bases ☒ coefficients that are the same.



Consider using the “Notice and Wonder” routine while students analyze and discuss the table from question 1. Since students are already asked to note their observations, consider challenging them to wonder about what this observation means in relation to the other properties of exponents learned previously.



MA.K12.MTR.5.1: Patterns and Structure
Probe students to take notice of the structure of using the laws of exponents. Position students to make connections between the expanded form and the form using the product of powers law.

2.1.2 Guided Instruction: Equivalent Algebraic Expressions (continued)

2. Complete the multiplication of $6b^7 \cdot 8b^{11}$ to find an equivalent monomial expression.

$$(6b^7)(8b^{11}) = (6 \cdot 8)(b^7 \cdot b^{11}) = 48b^{18}$$



Product of Powers

$a^m \cdot a^n = a^{m+n}$ and conversely $a^{m+n} = a^m \cdot a^n$

3. Determine an equivalent monomial algebraic expression by filling in the blanks.

$$(3a^3b^2)(7a^{12}b) = (3 \cdot 7)(a^3 \cdot a^{12})(b^2 \cdot b) = 21a^{15}b^3$$

4. Abigail and Mia both attempted to write an expression equivalent to $30x^7$.

Abigail	Mia
$30x^7 = (6x^2)(5x^5)$	$30x^7 = (3x^4)(10x^3)$

Discuss with your partner whether or not both students generated equivalent expressions. Write a summary of your discussion.

Answers will vary. Yes, both students generated equivalent expressions equal to $30x^7$. With Abigail it is $6 \times 5 = 30$ and $x^2 \times x^5 = x^{2+5} = x^7$ and Mia's work is $3 \times 10 = 30$ and $x^4 \times x^3 = x^{4+3} = x^7$.



When modeling how to multiply two binomials, consider that some students will benefit from seeing the expanded form of the expressions to see why the laws of exponents are being applied.



Provide students the opportunity to independently read, process, and answer the prompt in question 4 before discussing with a partner. Think-Pair-Share gives students three opportunities to make sense of information, first for independent learners, then pairing for verbal processing in order to give opportunities for more confident, precise responses when students share out.

2.1.3 Guided Practice: Products of Monomials

Component Overview: Students engage in four practice problems where they are asked to generate equivalent algebraic monomial expressions. Monitor students as they work and use questioning if students need help finding an entry point to a problem.



2.1.3 Guided Practice: Products of Monomials

1. Generate an equivalent algebraic monomial expression.

a. $(-9t^0u^4)\left(\frac{1}{3}u^{11}\right)(-10t^{20}u^8)$
 $30t^{20}u^{23}$

b. $(9q^{-3})(0.6q^{10})(12q^5)$
 $64.8q^{12}$

c. $(-4x^0y^2)(-10x^3y^5)(1.2x^4)$
 $48x^7y^7$

d. $(1.2m^2n^3)(-2.25m^7)(-0.5n^{12})$
 $1.35m^9n^{15}$



Although the answers to questions 1b and 1c end with positive exponents, this is a good opportunity to spiral content regarding the negative exponent law and the zero exponent law, which will be applied throughout the unit.



Consider challenging students who finish early to create an Error Analysis problem for others to analyze and solve. Creating Error Analysis problems positions students to think deeply about the content and the common mistakes and errors that occur.



Consider using different methods of presentation to best meet the students individually. Consider providing 3 options based on observations from previous lessons.

- Partner work
- Individual assignment
- Whole-group instruction

2.1.4 Collaboration: Raising a Monomial to a Power

Component Overview: Students work collaboratively with a partner to solve, analyze, and generate equivalent algebraic expressions. Monitor students as they work in pairs and use assessing and advancing questions for students who need help finding an entry point.



2.1.4 Collaboration: Raising a Monomial to a Power

Additional laws of exponents can be applied to generate equivalent algebraic expressions.



Power of a Product

$(ab)^m = a^m \cdot b^m$ and conversely $a^m \cdot b^m = (ab)^m$



Power of a Power

$(a^m)^n = a^{m \cdot n}$ and conversely $a^{m \cdot n} = (a^m)^n$

1. Two students began to generate an equivalent expression to $(-2x^4y)^3$.
 - a. The beginning of each student's work is shown. Finish each student's work.

Kaniya's Work	$(-2x^4y)^3$ $(-2)^3(x^4)^3(y)^3$ $= -8x^{12}y^3$
Nicholas' Work	$(-2x^4y)^3$ $(-2x^4y)(-2x^4y)(-2x^4y)$ $(-2 \cdot -2 \cdot -2)(x^4 \cdot x^4 \cdot x^4)(y \cdot y \cdot y)$ $= -8x^{12}y^3$



Students are positioned to compare different approaches to generate equivalent expressions. Utilize the structure of the lesson to encourage students to compare the methods by each student and connect that approach to their own understanding of the laws of exponents.



How do the power of a product and power of a power properties relate to your previous knowledge when multiplying monomials and using the product of powers property?

2.1.4 Collaboration: Raising a Monomial to a Power (continued)

- b. Discuss with your partner how Kaniya's and Nicholas' approaches to generating an equivalent expression were different but resulted in the same answer. Write a summary of your discussion.

Kaniya applied the exponent to each component of the term, while Nicholas multiplied the entire term by itself three times.

2. Use the expression $(12r^5)^2 \cdot \left(\frac{1}{4}r^7\right)$ to complete the following.

- a. Find an equivalent expression. Show your work.

$$\begin{aligned} & (12r^5)(12r^5) \left(\frac{1}{4}r^7\right) \\ &= (144r^{10}) \left(\frac{1}{4}r^7\right) \\ &= (144) \left(\frac{1}{4}\right) (r^{10})(r^7) \\ &= 36r^{17} \end{aligned}$$

- b. With your partner, compare and discuss the methods you used to create an equivalent algebraic expression.

Answers will vary. I simplified the first term first and then multiplied it by the second term while my partner did the coefficients first and then the exponents.



Consider asking the following after question 2b.

- How did your approach differ from your partner's?
- Did you both generate the same expression? If not, who has the correct expression?
- Is there only one correct answer?

2.1.5 Your Turn!

Component Overview: Students work independently to determine equivalent monomial expressions. Use assessing and advancing questions if needed to position students to determine the method needed to solve.



2.1.5 Your Turn!

- Determine an equivalent monomial expression for each expression using the laws of exponents. Show your work.

$\begin{aligned} &(-3w^3)^2(20z^9)(-7z^{14}) \\ &(-3)^2(w^3)^2(20)(z^9)(-7)(z^{14}) = \\ &(-3)^2(20)(-7)(w^3)^2(z^9)(z^{14}) = \\ &(9)(20)(-7)(w^6)(z^9)(z^{14}) = \\ &-1,260w^6z^{23} \end{aligned}$	$\begin{aligned} &(23k^6)^0\left(\frac{2}{3}k\right)(k^{14}) \\ &(23)^0(k^6)^0\left(\frac{2}{3}\right)(k)(k^{14}) = \\ &(23)^0\left(\frac{2}{3}\right)(k^6)^0(k)(k^{14}) = \\ &(1)\left(\frac{2}{3}\right)(k^0)(k)(k^{14}) = \\ &\frac{2}{3}k^{15} \end{aligned}$
$\begin{aligned} &(27b^{30})\left(\frac{1}{3}b^9\right) \\ &(27)\left(\frac{1}{3}\right)(b^{30})(b^9) = \\ &9b^{39} \end{aligned}$	$\begin{aligned} &(6q^3s^7)^2(-q^2s^{-1})(-7q^{10}s^{14}) \\ &(6)^2(q^3)^2(s^7)^2(-1)(q^2)(s^{-1})(-7)(q^{10})(s^{14}) \\ &= (6)^2(-1)(-7)(q^3)^2(q^2)(q^{10})(s^7)^2(s^{-1})(s^{14}) \\ &= (36)(-1)(-7)(q^6)(q^2)(q^{10})(s^{14})(s^{-1})(s^{14}) \\ &= 252q^{18}s^{27} \end{aligned}$
$\begin{aligned} &(-11bc^4)^3 \\ &(-11)^3(b)^3(c^4)^3 = \\ &(-11)^3(b)^3(c^{12}) = \\ &-1,331b^3c^{12} \end{aligned}$	$\begin{aligned} &(-4x^2y^{12})^{-2} \\ &\frac{1}{(-4x^2y^{12})^2} = \\ &\frac{1}{(-4)^2(x^2)^2(y^{12})^2} = \\ &\frac{1}{(-4)^2(x^4)(y^{24})} = \\ &\frac{1}{16x^4y^{24}} \end{aligned}$



Throughout the activity, consider asking the following to advance students through the work.

- How can the order of operations be used to determine where to start?
- What is the exponent attached to?
- Which method are you using to solve (expanded or properties)?
- How might the expanded form help you determine the equivalent expression?
- What do the laws of exponents apply to? Does that include coefficients?



Challenge students to work problems both using the laws of exponents and the expanded form. For example, encourage students to work three problems using the properties and three problems using the expanded form to demonstrate conceptual understanding of both approaches.



Consider providing students with choice as to which problems they complete. Instead of positioning students to answer all six questions, consider having students choose four of the six questions to complete.

2.1.6 Wrap-Up: Cool Down

Component Overview: Consider using all or parts of this exercise as a formative assessment to gather individual student data on student progress from this lesson. This Wrap-Up assesses student understanding of determining equivalent monomial expressions using the product laws of exponents.



2.1.6 Wrap-Up: Cool Down

Determine an equivalent expression for each algebraic expression using the laws of exponents. Show your work.

1. $(56g^7)\left(\frac{4}{5}g^{-2}\right)$

$$(56)\left(\frac{4}{5}\right)(g^7)(g^{-2}) = \frac{224}{5}g^5$$

2. $(-x^2y)(7.3x^8y^{11})(0.1x^9y^3)^2$

$$\begin{aligned} &(-1)(x^2)(y)(7.3)(x^8)(y^{11})(0.1)^2(x^9)^2(y^3)^2 = \\ &(-1)(7.3)(0.1)^2(x^2)(x^8)(x^9)^2(y)(y^{11})(y^3)^2 = \\ &(-1)(7.3)(0.01)(x^2)(x^8)(x^{18})(y)(y^{11})(y^6) = \\ &-0.073x^{28}y^{18} \end{aligned}$$